



DEMO VIDEO SCRIPT

VayuNetra

The 3-minute demo, beat by beat

Total runtime 3:00 — three presenters, one minute each. The split below (**Omkar → Abhinav → Sejal**) is a suggested order following the product loop trace → act → protect; any of the three can take any segment. Every beat has a timestamp, the exact screen, and the exact words.

Team Omkar Kadam · Sejal Kumbhar · Abhinav Prasad

Event ET AI Hackathon 2026 · Problem Statement 5

Live vayunetra-aqi.vercel.app · **API** vayunetra-c8i8.onrender.com · 20 July 2026

The spoken lines are written for natural Indian-English delivery at ~140 words/minute — read them aloud twice before recording; if a line feels stiff in your mouth, reword it, but **keep every number exactly as written** (all are verified against the live app; see the note in Minute 2 about re-checking on record day).

BEFORE ANYONE RECORDS

Recording setup — do this once

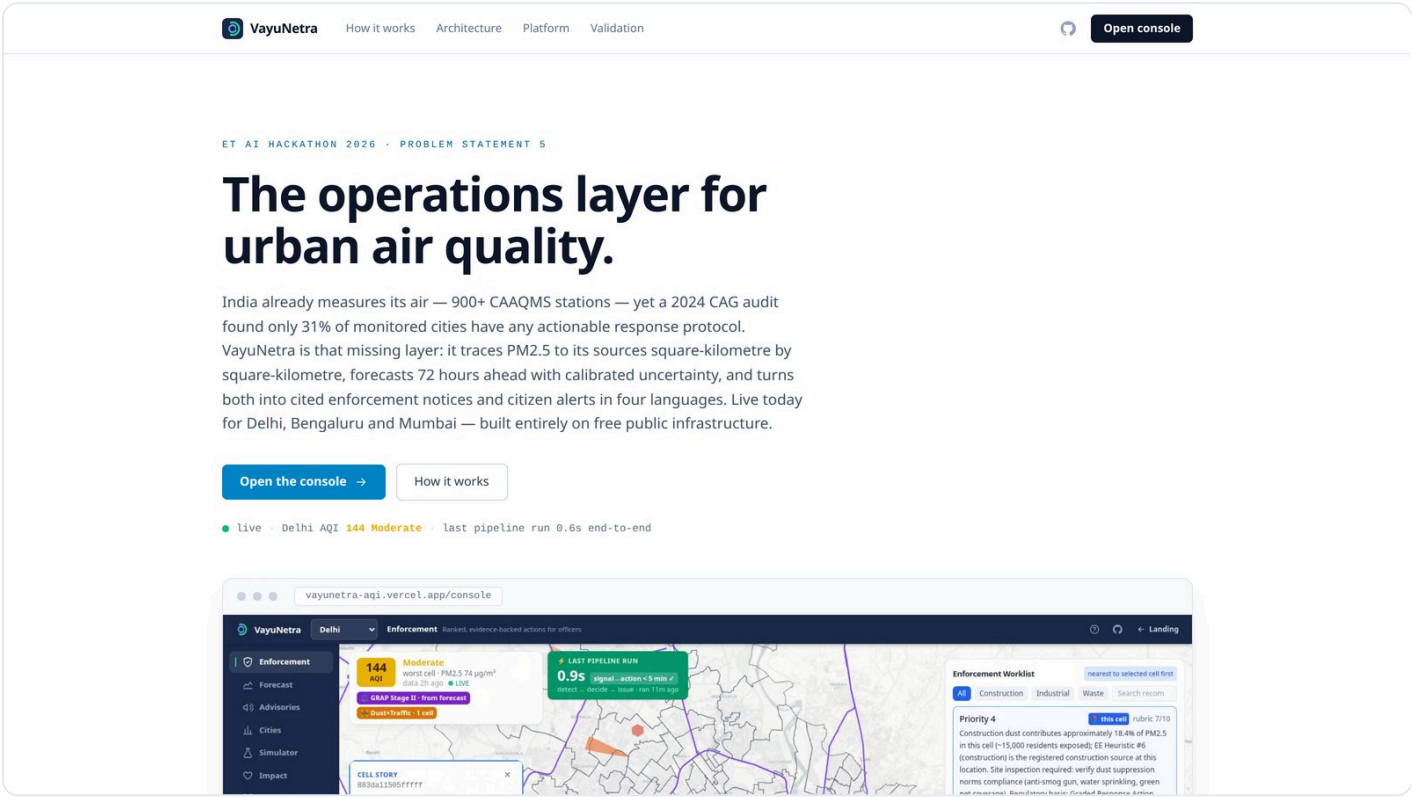
Item	Setting
Screen capture	1920×1080, browser full-screen (F11), 100% zoom, hide the bookmarks bar
Camera	Each speaker records a webcam clip; the editor overlays it as a small circle, bottom-left (never cover the right panel — that's where the content lives)
Audio	Quiet room, phone/headset mic 15 cm from mouth, record 5 seconds of silence first for noise removal
Browser prep	Open <code>vayunetra-aqi.vercel.app</code> once and complete/skip the tour so it never pops up on camera. City = Delhi .
Backend prep	Run <code>make prewarm</code> ~15 min before recording and confirm GO . If Render misbehaves, record against local <code>make dev</code> — the app looks identical.
Cursor	Move slowly, click deliberately. Every click in this script is rehearsed — no hunting on camera.

Pre-record checklist (2 min): ① tour dismissed ② Delhi selected ③ a hexagon click opens Cell Story instantly ④ Enforcement list is populated ⑤ "Run agents live" completes in a few seconds ⑥ Advisories shows Hindi text ⑦ Sejal's phone has Telegram open on **@aqivayu_bot** for the pickup shot.

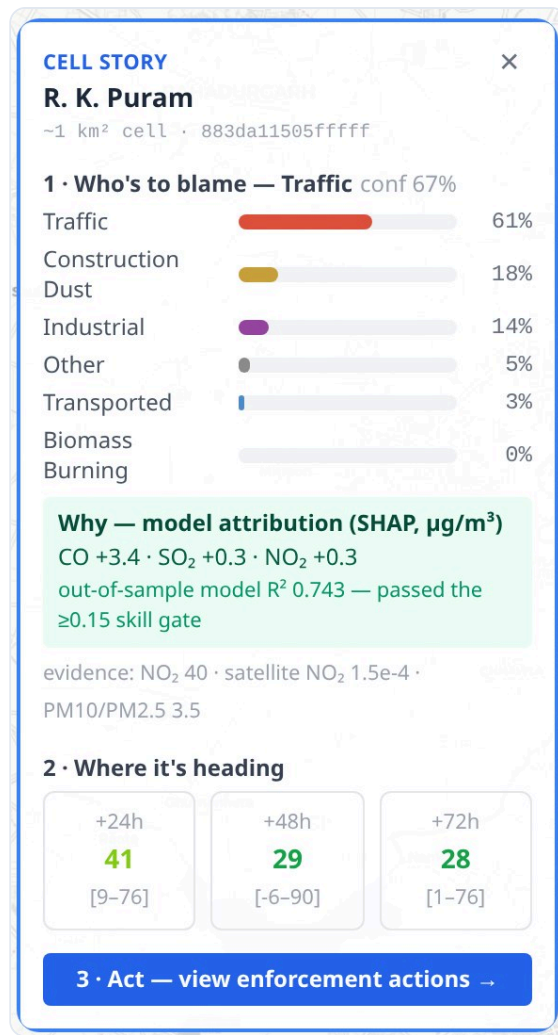
Minute 1 — OMKAR · "Who is to blame, per square kilometre"

Time	Screen / action	Say exactly
0:00–0:10	Landing page (<code>vayunetra-aqi.vercel.app</code>), slow scroll over the hero and the live AQI line	"India already measures its air — nine hundred monitoring stations. Yet only thirty-one percent of monitored cities have any response protocol. Data is not the problem. The missing loop is."
0:10–0:22	Click Open console → map of Delhi loads with coloured hexagons. Sweep cursor across the map	"This is VayuNetra. Every hexagon is one square kilometre of Delhi, coloured by who is to blame for its PM2.5 right now — red is traffic, yellow is construction dust, purple is industry."
0:22–0:40	Click a hexagon → Cell Story opens (place name, e.g. R. K. Puram). Point at the source bars, then the green SHAP box	"Click any cell — this is R. K. Puram. Traffic: sixty-one percent. And the model shows its evidence — SHAP drivers in micrograms, with its own out-of-sample R-squared on display. Where the model has no skill, it abstains and falls back to cited chemical signatures. It never over-claims."
0:40–0:55	Point at "Where it's heading" (+24/48/72h numbers), then click Forecast in the sidebar; hover the uncertainty band	"The same cell carries a seventy-two-hour forecast with calibrated eighty-percent uncertainty bands. Our attribution matches official inventories at cosine zero-nine-two, and the forecast beats persistence in all three cities — we validated everything, including our failures."
0:55–1:00	Hold on the Forecast chart	"So we know who is to blame, and what's coming. Abhinav turns that into action."

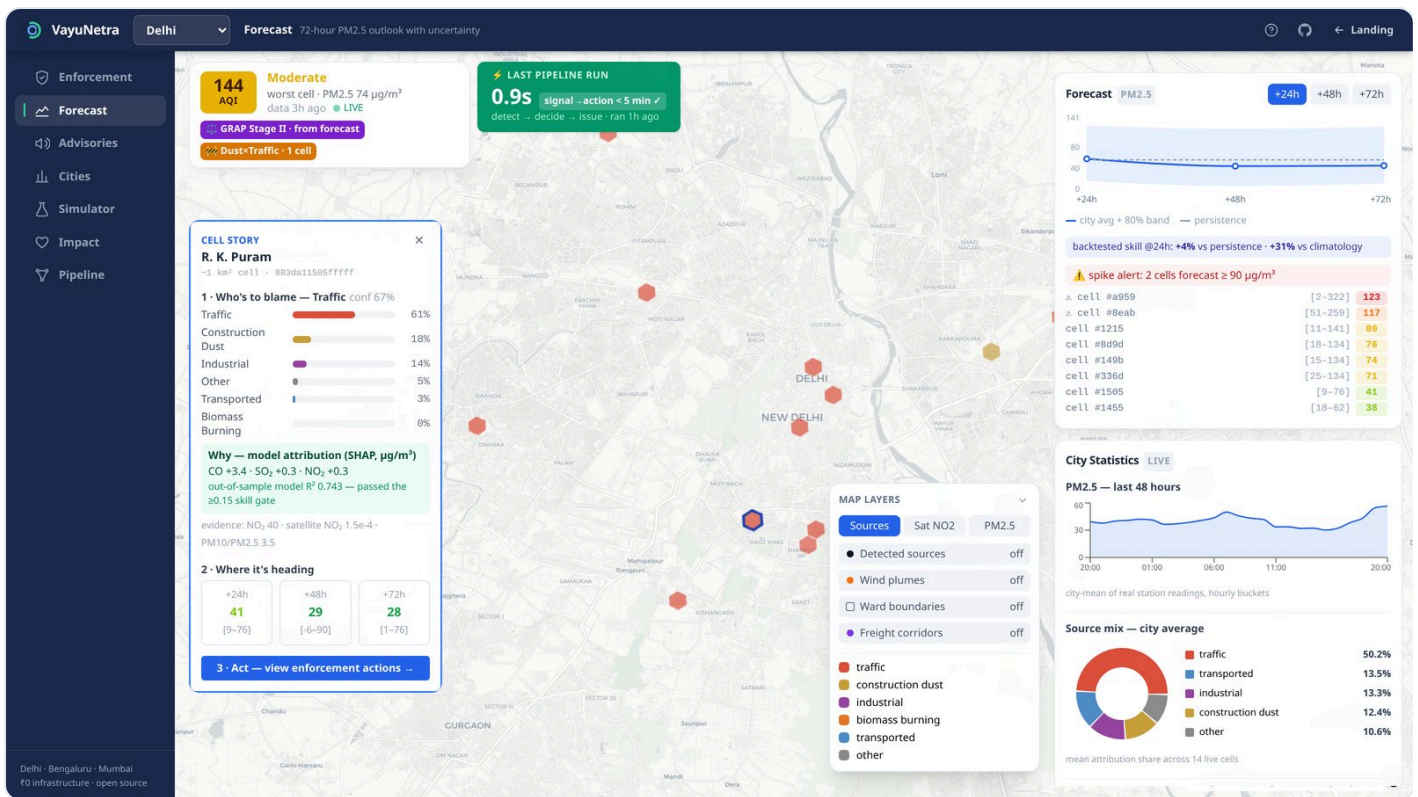
~150 words of speech. If running long, drop "we validated everything, including our failures."



0:00 — the opening frame. The landing hero with the live Delhi AQI status line. Scroll slowly; don't click anything until "The missing loop is."



0:22 — the Cell Story panel after clicking a hexagon: place name (R. K. Puram), source-share bars (Traffic 61%), the green SHAP evidence box with the model's R², and the 72-hour outlook. Point at each as you speak.



0:40 — the Forecast section. Hover the shaded band while saying "calibrated eighty-percent uncertainty bands"; the gray dashed line is the persistence baseline you "beat".

Minute 2 — ABHINAV • "From blame to a signed notice"

Time	Screen / action	Say exactly
1:00–1:15	Click Pipeline in the sidebar → click " Run agents live " → the trace fills in node by node	<i>"Six AI agents on one LangGraph run this city. Watch them live: orchestrator, attribution, forecast — a spike gate decides if the air needs enforcement — then advisory. Zero-point-six seconds end to end, and every node is traced and auditable."</i>
1:15–1:32	Click Enforcement . Slowly scroll the worklist; point at the contribution + exposed line on the top card	<i>"The enforcement worklist ranks every polluting source by contribution, people exposed, and actionability. Top of this list: a registered construction site putting eighteen percent of this cell's PM2.5 onto fifteen thousand residents."</i>
1:32–1:50	Click Evidence dossier on that card → point at the Sentinel-2 patch and the citations. Then click Notice PDF → open the download, scroll to the impact chart	<i>"One click — the evidence dossier: a Sentinel-2 satellite patch of the site, and regulatory citations retrieved from GRAP and the Air Act. And here's the draft notice PDF — with a projected-impact chart: the forecast if nothing is done, versus the air after compliance. An officer signs it. We never auto-send."</i>
1:50–2:00	Back in the app, show the Intervention tracking card under the worklist	<i>"The moment it's dispatched, the system freezes a baseline and starts measuring before-versus-after, corrected for city drift. And Sejal takes the same intelligence to citizens."</i>

~150 words. Pre-download the Notice PDF once so the on-camera download is instant.

Numbers on record day: "eighteen percent ... fifteen thousand residents" matches the live top card when a cell is focused (verified 20 Jul). The worklist re-ranks with live data — before recording, read your actual top card and, if it differs, swap in its contribution % and exposed count. The sentence structure stays the same; **never speak a number that isn't on screen.**

Agent Pipeline

0.6s end-to-end

last multi-agent run · detect → decide → issue

	Orchestrator	+0.0s	
	Attribution	+0.2s	0.2s
	Forecast	+0.4s	0.2s
	Spike gate	no spike → straight to advisory	
	Enforcement	skipped — air is clean, nothing to enforce	
	Advisory	+0.6s	0.2s
	Multi-City	cross-city playbooks — see the Cities section	

► Run agents live

1:00 — the Agent Pipeline card after "Run agents live": all six agents with per-node timings, the spike-gate decision, and the end-to-end badge (0.6 s here). Let it fill in on camera — it takes about a second.

VayuNetra Delhi Enforcement Ranked, evidence-backed actions for officers

144 AQI Moderate worst cell: PM2.5 74 $\mu\text{g}/\text{m}^3$ data 4h ago LIVE

GRAP Stage II - from forecast

Dust-Traffic: 1 cell

LAST PIPELINE RUN 0.6s signal - action < 5 min detect - decide - issue - ran 25m ago

CELL STORY R. K. Puram ~1 km² cell - 883da11505fffff

1 - Who's to blame - Traffic conf 67%

Traffic 61% Construction 18% Industrial 14% Other 5% Transported 3% Biomass Burning 0%

Why - model attribution (SHAP, $\mu\text{g}/\text{m}^3$) CO +3.4 SO₂ +0.3 NO₂ +0.3 out-of-sample model R² 0.743 - passed the ≥ 0.15 skill gate

evidence: NO₂ 40 - satellite NO₂ 1.5e-4 PM10/PM2.5 3.5

2 - Where it's heading

+24h 41 [9-76] +48h 29 [4-90] +72h 28 [1-76]

3 - Act - view enforcement actions

MAP LAYERS Sources Sat NO2 PM2.5

Detected sources off Wind plumes off Ward boundaries off Freight corridors off

traffic construction dust industrial biomass burning transported other

Enforcement Worklist nearest to selected cell first

All Construction Industrial Waste Search recom

Priority 4 this cell rubric 7/10

Construction dust contributes approximately 18.4% of PM2.5 in this cell (~15,000 residents exposed); EE Heuristic #6 (construction) is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

18% contribution - 15,000 exposed +11 similar sites here

Hide dossier Notice PDF

SATELLITE EVIDENCE

Sentinel-2 patch - EE Heuristic #6 (construction) construction - 69% detection confidence

REGULATORY CITATIONS (RAG)

Graded Response Action Plan (GRAP) match 24%

hospitals: additional respiratory care wards activated. - Helpline numbers published; mobile health units deployed.

Priority 1 ~6 km rubric 6/10

Construction dust contributes approximately 9.2% of PM2.5 in this cell (~15,000 residents exposed); EE Heuristic #40

1:32 — the expanded evidence dossier on the top worklist card: the real Sentinel-2 satellite patch with detection confidence, and the GRAP citation with its match score. This is the "one click" moment.

VAYUNETRA Urban Air Quality Intelligence - Enforcement Cell DRAFT FOR OFFICER REVIEW

ENFORCEMENT NOTICE

REFERENCE VN-ENF-0341

DATE 19 July 2026

STATUS DRAFT - pending officer authorisation

PREPARED BY VayuNetra decision-support system

TO The Occupier / Site Manager, EE Heuristic #6 (construction), Delhi (grid cell 883da11505fffff)

SUBJECT Non-compliance with air pollution control norms - construction dust

FINDINGS Construction dust contributes approximately 18.4% of PM2.5 in this cell (~15,000 residents exposed); EE Heuristic #6 (construction) is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

EXPOSURE ASSESSMENT An estimated 15,000 residents live within the affected ~1 sq. km grid cell (883da11505fffff). VayuNetra source attribution assigns approximately 18.4% of the local PM2.5 concentration to this source (GPW v4 population x model attribution).

APPLICABLE REGULATIONS

- Graded Response Action Plan (GRAP)

SATELLITE EVIDENCE Sentinel-2 patch - EE Heuristic #6 (construction) (69% detection confidence)

VayuNetra AI - system-generated draft. Verify and authorise before official issuance. Page 1

1:40 — Notice PDF, page 1: addressee, cited provisions, 24-hour IST deadline.

Sentinel-2 patch - EE Heuristic #6 (construction)

The image above forms part of the digital evidence dossier for this recommendation and is available in the VayuNetra console.

PROJECTED IMPACT OF COMPLIANCE Modeled linear rollback of this source's 18.4% share on the cell's central PM2.5 forecast ($\mu\text{g}/\text{m}^3$) - a screening estimate, not a guarantee.

$\mu\text{g}/\text{m}^3$

41 34 29 24 28 23

+24h +48h +72h

forecast, no action with source compliance (modeled)

+24h: 41.1 -> 33.5 - +48h: 28.9 -> 23.6 - +72h: 28.3 -> 23.1

REQUIRED ACTION The occupier shall undertake immediate corrective measures and demonstrate compliance by 21 July 2026, 00:46 IST (24 hours from issue). Continued non-compliance may attract penalties and/or site sealing under the applicable provisions.

REPRESENTATION A written representation, with supporting documents, may be submitted to the authorising officer within the compliance window.

AUTHORISATION

Name: _____ Designation: _____

Signature: _____ Date: _____

This is a system-generated draft for officer review before issuance.

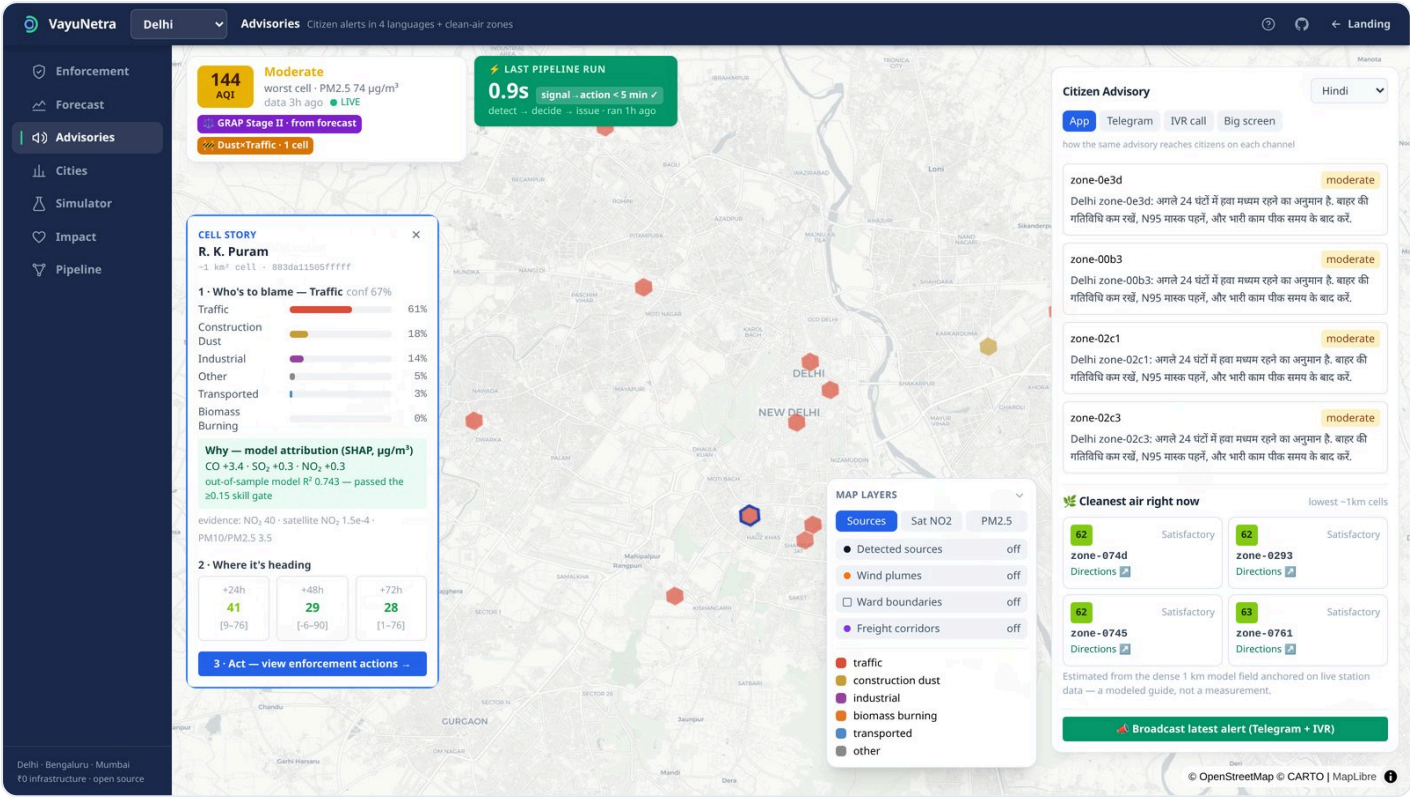
VayuNetra AI - system-generated draft. Verify and authorise before official issuance. Page 2

1:45 — Notice PDF, page 2: satellite evidence and the projected-impact chart — pause on the red-vs-green bars.

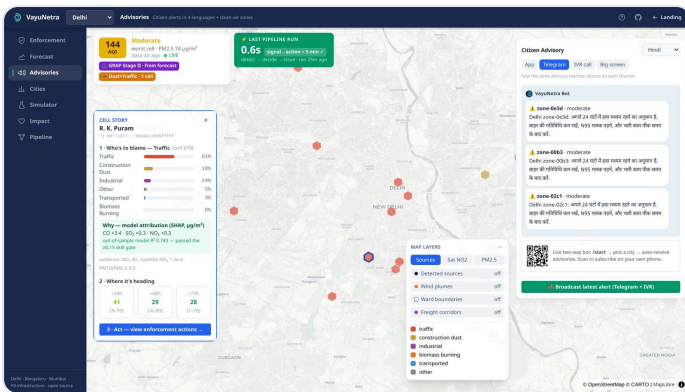
Minute 3 — SEJAL · "Reach, rupees, and the close"

Time	Screen / action	Say exactly
2:00–2:18	Click Advisories . Switch the language dropdown English → Hindi (Devanagari appears). Click through channel tabs App → Telegram → IVR call	"The same intelligence reaches people who will never open a dashboard. Health advisories in English, Hindi, Kannada and Marathi — targeted using two thousand six hundred vulnerability-scored zones: hospitals, schools, outdoor workers."
2:18–2:32	On the Telegram tab, show the QR; cut to Sejal's phone receiving the bot message. Then the IVR tab	"Four channels. A live Telegram bot — scan and subscribe right now. Public display boards. And real IVR phone calls: citizens dial in, press one for Delhi, and the phone reads them today's advisory."
2:32–2:48	Click Simulator → intervention " Halt construction dust " → Run simulation → point at the impact cards. Then click Impact , point at the ₹ figures	"Officers can ask what if: halt construction dust, and the counterfactual returns the AQI change, people protected, rupees saved and CO2 avoided — every figure cited, nothing invented. The City ROI view makes Delhi's funding case: eighty-eight thousand crore avertable every year."
2:48–3:00	Click Cities (three-city comparison) — then cut to the landing hero, hold two seconds	"Three cities run today — city number four is one YAML file away, on zero-rupee infrastructure. India's air stack can measure and forecast. VayuNetra makes it operate. Thank you."

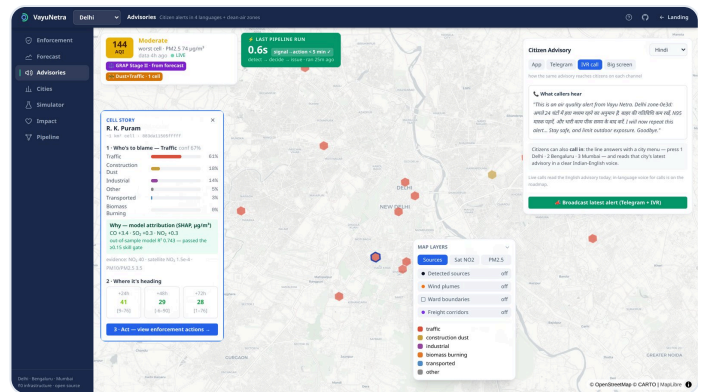
~145 words. The phone-pickup shot is the single most memorable moment — rehearse it.



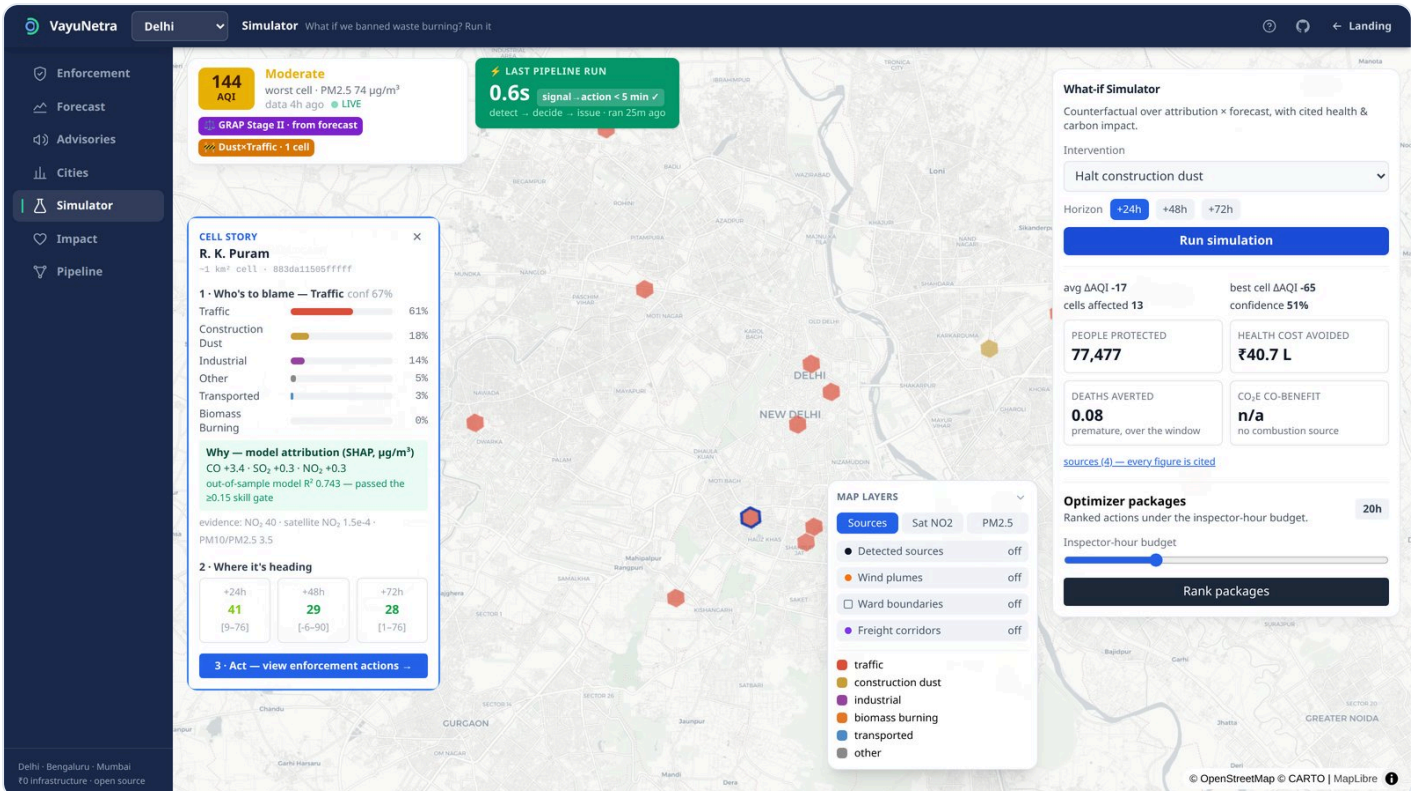
2:00 — **Advisories in Hindi** (App tab): ward-level advisories in Devanagari, plus the "Cleanest air right now" zones below. The language dropdown is top-right of the panel.



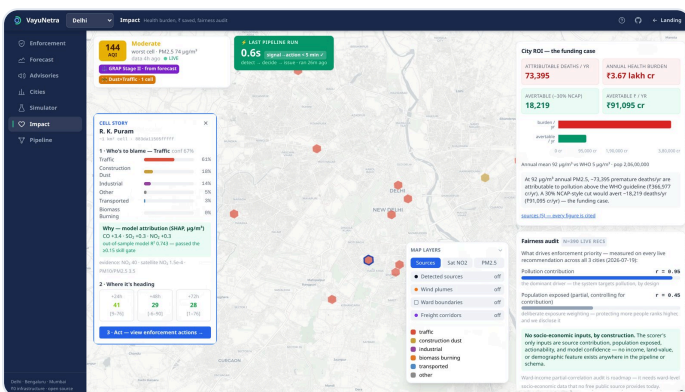
2:18 — Telegram tab: bot preview + the QR to scan on camera.



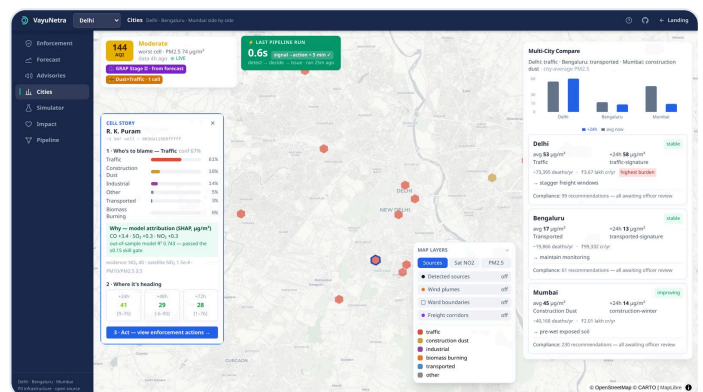
2:25 — IVR tab: the spoken script and the call-in menu ("press 1 Delhi...").



2:32 — the Simulator after a run ("Halt construction dust", +24h): Δ AQI, people protected, health cost avoided — and the honest "n/a — no combustion source" CO₂e card. Point at the cards left to right.



2:40 — Impact: the City ROI funding case with the ₹ figures to point at.



2:48 — Cities: the three-city comparison for the final beat before the landing-page close.

Coverage check — nothing missed

Landing ✓ (0:00) · Blame map ✓ (0:10) · Cell Story: attribution, SHAP, abstention, forecast ✓ (0:22–0:55) · Forecast panel ✓ (0:40) · Pipeline: 6 agents, spike gate, latency ✓ (1:00) · Enforcement worklist ✓ (1:15) · Dossier: satellite + RAG citations ✓ (1:32) · Notice PDF + impact chart ✓ (1:32) · Intervention tracking ✓ (1:50) · Advisories: 4 languages ✓ (2:00) · 4 channels + live Telegram + IVR ✓ (2:18) · Simulator + cited impact ✓ (2:32) · Impact / City ROI ✓ (2:32) · Cities comparison ✓ (2:48) · Scalability + ₹0 close ✓ (2:48).

Not shown (deliberate, no time): guided tour, map layer toggles (plumes/wards/freight are visible in the landing hero image), clean-air zones, optimizer, broadcast button. If a judge asks — all are live in the app.

Editing notes

- Hard cuts on every speaker change; no transitions or music under speech (a light music bed at –25 dB is fine).
- Overlay small text chips when a number is spoken:

61% traffic — 0:25

0.6 s end-to-end — 1:10

18% · 15,000 residents — 1:25

en · hi · kn · mr — 2:05

₹0.88 L cr / yr — 2:45
- End-card (2 s, after 3:00 if rules allow a title card): logo + `vayunetra-aqi.vercel.app` + team names.
- Export 1080p, H.264, target ≤ 500 MB.